

Relative Survival Rates after Alternative Therapies for Uveal Melanoma

JOHANNA M. SEDDON, MD,^{1,2} EVANGELOS S. GRAGOUDAS, MD,²
KATHLEEN M. EGAN, MPH,¹ ROBERT J. GLYNN, ScD,¹ SUZANNE HOWARD, BS,¹
ROBERT G. FANTE, MD,¹ DANIEL M. ALBERT, MD³

Abstract: Survival in a group of 556 patients with uveal melanoma treated by proton beam irradiation with a median follow-up of 5.3 years was compared with that of 238 patients enucleated during the same 10-year period as irradiated patients (July 1975 to December 1984) with a median follow-up of 8.8 years, and 257 patients enucleated during the preceding 10 years (January 1965 to June 1975) with a median follow-up of 17.0 years. Adjustments were made for known prognostic factors including age, tumor location, tumor height, and clinical estimate of tumor diameter (for enucleated patients this was estimated in a regression equation relating histologic to clinical measurement). The overall rate ratio for all cause mortality was 1.2 (95% confidence interval, 0.9–1.6) for the concurrent enucleation series versus proton beam, and 1.6 (95% confidence interval, 1.2–2.1) for the earlier enucleation series versus proton beam. Relative rates of metastatic death, cancer death, and all cause mortality comparing alternative treatments were found to vary with time after treatment. Interval-specific rate ratios were evaluated using proportional hazards models fitted to separate time intervals after treatment. For all three outcomes, rate ratios were over two and statistically significant for the first 2 years after treatment and closer to one and nonsignificant after year 6 comparing the two enucleation groups with proton beam. Results suggest that treatment choice has little overall influence on survival in patients with uveal melanoma.

Ophthalmology 1990; 97:769–777

Uveal melanoma threatens sight as well as life. Methods to preserve sight are well known, but prolongation of life is still a challenge. Several treatment modalities are avail-

able for uveal melanoma including enucleation, enucleation preceded by conventional irradiation, proton beam irradiation, I^{125} , cobalt plaque, and $^{106}\text{Ru}/^{106}\text{Rh}$ applicator. A randomized clinical trial is underway which is comparing survival rates of patients after three of these treatments: enucleation, enucleation with preoperative radiation, and I^{125} .¹ Results are expected in several years.

Patients seeking care for this potentially life-threatening disease expect answers to their questions concerning benefits and risks of treatment, based on current evidence. Even though it would be preferable to know the effect of treatments on survival based on randomized trials, there is still a need to provide the best assessment available for both patients and their physicians. To this end, survival patterns in the first few years after treatment with proton irradiation or enucleation were evaluated with the finding that survival after proton beam was not worse than survival after enucleation in the early years after treatment.²

Originally received: October 29, 1989.

Revision accepted: January 22, 1990.

¹ Epidemiology Unit, Department of Ophthalmology, Massachusetts Eye and Ear Infirmary, Harvard Medical School, Boston.

² Retina Service, Department of Ophthalmology, Massachusetts Eye and Ear Infirmary, Harvard Medical School, Boston.

³ David G. Cogan Ophthalmic Pathology Laboratory, Department of Ophthalmology, Massachusetts Eye and Ear Infirmary, Harvard Medical School, Boston.

Presented at the American Academy of Ophthalmology Annual Meeting, New Orleans, Oct/Nov 1989.

Reprint requests to Johanna M. Seddon, MD, Massachusetts Eye and Ear Infirmary, 243 Charles St, Boston, MA 02114.